

Melnikov's valency-variety problem

OPG-46575

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

Problem The valency-variety $w(G)$ of a graph G is the number of different degrees in G . Is the chromatic number of any graph G with at least two vertices greater than

$$\left\lceil \frac{\lfloor w(G)/2 \rfloor}{|V(G)| - w(G)} \right\rceil ?$$